CNC Mill: Operation and Maintenance Manual

Applies to: DMG Mori DMU 50, Haas VF-2, Mazak VCN-530C. Document no. OM-CNC-MILL-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
temperature	spindle bearing	degC	approx. 55	> 75	> 90
vibration	spindle housing	mm/s	approx. 2	> 4.5	> 7.1
rpm	spindle	rpm	approx. 8000	> 11000	> 12000
current	main drive	A	approx. 18	> 28	> 34

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High temperature (spindle bearing, degC)

Probable causes:

- Insufficient oil-air lubrication of the spindle bearings
- Spindle chiller fault or low coolant level
- Excessive bearing preload after a tool crash
- Prolonged operation above 10,000 rpm

Corrective actions:

1. Check the oil-air lubrication unit: supply pressure must be 4-6 bar.
2. Verify the spindle chiller set point is 20 +/- 2 degC and the coolant level is above MIN.
3. Run the spindle warm-up program (50% speed, 10 min) before production.
4. At HIGH_HIGH: stop the spindle, let it cool for 30 min and inspect the bearings before restart. Replace with bearing set SP-CNC-BRG-7014 if damaged.

4.2 High vibration (spindle housing, mm/s)

Probable causes:

- Tool or tool-holder imbalance

- Worn spindle bearings
- Low drawbar clamping force
- Chatter from aggressive cutting parameters

Corrective actions:

1. Balance tools to grade G2.5 at 12,000 rpm.
2. Measure drawbar clamping force: minimum 18 kN; replace disc springs (SP-CNC-DSP-018) if lower.
3. Reduce depth of cut by 30% and re-check.
4. Above 7.1 mm/s (ISO 10816-3 zone D) stop the machine and replace bearing set SP-CNC-BRG-7014.

4.3 High rpm (spindle, rpm)

Probable causes:

- Program S-word above the rated spindle speed
- Spindle encoder cable damaged
- Drive parameter set incorrectly after service

Corrective actions:

1. Check the NC program speed command; rated maximum spindle speed is 12,000 rpm.
2. Inspect the encoder cable and connector X12 on the spindle drive.
3. Restore drive parameters from the last backup (parameter group P1000).

4.4 High current (main drive, A)

Probable causes:

- Worn or dull cutting tools
- Feed rate too high for the material
- Insufficient axis way lubrication

Corrective actions:

1. Check tool wear; replace inserts when flank wear exceeds 0.3 mm.
2. Reduce feed override by 20% and observe the drive current.
3. Check the way-lube tank level and the lubrication interval (every 30 min of axis travel).

5. Maintenance schedule

Interval	Task
Daily	Check coolant concentration (6-8%) and level
Weekly	Clean chip conveyor; check way-lube level
Every 30 days	Spindle lubrication service and filter change
Every 6 months	Spindle runout check (max 0.003 mm)
Yearly	Ball-screw backlash compensation and geometry check